Douwe Rijs

Robotics & Machine Learning Engineer · MSc Robotics, TU Delft

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Robotics MSc student at TU Delft with a Mechanical Engineering background and a strong focus on machine learning, perception and control systems. Hands-on R&D experience in computer vision, transformer models and enterprise software development at Allseas. Motivated to apply advanced robotics and ML to real-world industrial systems.

EXPERIENCE

Future Flight Operations Intern — Boeing Innovation Lab, Neu-Isenburg (Germany) Sep – Dec 2026 (upcoming)

Mandatory MSc internship on autonomy and AI/ML applications for future flight operations.

Assistant R&D Engineer — Allseas, Delft Feb 2025 – Aug 2026

- Built a network observability platform polling 100+ switches via SNMP, aggregating port-level metrics (data rate, VLAN, link status) for real-time diagnostics.
- Designed an R&D computer-vision/VLM pipeline for reading pipe IDs on curved steel — see Selected Projects.
- Implemented Azure DevOps CI/CD pipelines automating versioning and software releases from commits.
- Day-to-day development in Go, React/TypeScript and Python within an enterprise software process.

EDUCATION

MSc Robotics — Delft University of Technology 2025 – present

Machine learning, deep reinforcement learning, machine perception (sensor fusion, Kalman filtering, LiDAR/radar/stereo vision), robot dynamics & control, motion planning (RRT, MPC).*

BSc Mechanical Engineering — Delft University of Technology 2022 – Oct 2025

Thesis (8.0/10): chemotherapy-injection assisting device — medical-device regulations & CE certification. Minor in Computer Science: ML fundamentals, camera geometry & 3D reconstruction, algorithms & software engineering.

VWO (pre-university) — Haarlemmermeerlyceum, Hoofddorp 2016 – 2022

SELECTED PROJECTS

Greenhouse Monitoring Robot & Digital Twin (MIRTE) — TU Delft, team of 5

- Autonomous ROS 2 robot mapping a greenhouse via SLAM (Nav2, SLAM Toolbox), inspecting flower beds and streaming plant/pest/climate data into a live digital twin.
- Designed and implemented a state machine for navigation, scanning, fault recovery and battery-triggered return-to-dock, with YOLOv8 detection and AprilTag localisation validated in simulation and on the real robot.
- Managed the team's GitLab workflow; 7 of 8 mandatory client requirements met in 8 weeks.

Autonomous Flight of a Micro Air Vehicle — TU Delft MAVLab

- Vision-only obstacle avoidance for a Parrot Bebop 2 flying autonomously in the CyberZoo using onboard compute only.
- Implemented border, gate and vertical-obstacle detection in C/C++ inside the Paparazzi autopilot; optimised the vision stack to 8–11 FPS on the drone's onboard ARM processor.

Pipe-ID Recognition Pipeline — Allseas R&D

- Built an end-to-end CV/VLM pipeline reading stencilled & chalked pipe IDs from photos taken inside pipeline pipes, raising exact-ID accuracy from ~1% to ~80%.
- Benchmarked four OCR engines (Tesseract, EasyOCR, PaddleOCR, TrOCR), then diagnosed failure modes and pivoted to a tuned VLM approach when classical OCR plateaued.
- Doubled exact-match accuracy (29% → 59%) purely through YOLO oriented-bounding-box crop preprocessing — before any model tuning — showing preprocessing outweighed model choice.
- Prototyped a stencil- and colour-agnostic ID locator using CRAFT text/linking scores with per-project arc-fitted text maps, generalising across pipelines with different layouts; evaluated 360° capture to remove blind-spot failures.

SKILLS

Programming	Python, Go, TypeScript/JavaScript, React, C++, MATLAB
Machine Learning	CNNs, RNNs, Transformers, VLMs, deep reinforcement learning, SVM, PCA, Hugging Face, YOLO
Robotics	ROS 2, SLAM, Nav2, sensor fusion, Kalman filtering, LiDAR, radar, stereo vision, pose estimation
Control & Planning	PID, MPC, impedance control, robot dynamics, RRT*, search algorithms
Tools & DevOps	Git, Azure DevOps, CI/CD, Docker, Linux, observability & monitoring
Languages	Dutch (native), English (fluent), Chinese (elementary)